

Justin M. Parra

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EDUCATION

Rutgers University – New Brunswick

New Brunswick, NJ

Bachelor of Science in Electrical and Computer Engineering (ECE)

Expected December 2027

- **Relevant Coursework:** Principles of Electrical Engineering I & II, Microelectronic Circuits, Linear Systems and Signals, Control Systems Principles, Feedback & State Estimation, Digital Logic Design, Computer Architecture, Embedded Systems & Microcontrollers.

TECHNICAL SKILLS

Languages: C, C++ (C++17), Python, MATLAB, Verilog HDL, Assembly, Bash, Java

Embedded & Microcontrollers: STM32 (HAL & LL, STM32F405), ESP32 / ESP32-S3, ARM Cortex-M4, FreeRTOS, Bare-Metal C, Timer ISRs, DMA, Arduino

Hardware & PCB Design: KiCad (4-Layer), Altium Designer, EasyEDA, 4-Layer PCB Routing, Controlled Impedance, Coplanar Ground Shielding, Analog Signal Conditioning, Active Op-Amp Filters, TVS Surge Protection, Decoupling

Protocols & Buses: I2C (400 kHz), SPI, UART, RS-485, CAN Bus, PWM Motor Control, GPIO, ADC/DAC

Test & Measurement: Digital Storage Oscilloscopes, Logic Analyzers, Digital Multimeters, Benchtop Debugging, Signal Analysis

Tools & Platforms: Linux (Arch, Ubuntu, WSL2), Git, GitHub CI/CD, CMake, Make, GDB, LTspice, SOLIDWORKS

ENGINEERING EXPERIENCE

Rutgers Micromouse

New Brunswick, NJ

Co-President & Lead Hardware/Firmware Engineer

September 2024 – Present

- Developed embedded firmware in **C/C++** on **STM32F405** and **ESP32-S3** microcontrollers for real-time motion control and sensor fusion utilizing **FreeRTOS**.
- Designed and routed custom **4-layer PCBs** in **KiCad** and **Altium Designer** with ENIG surface finish, integrating **STM32F405**, IMU, optical ToF sensors, and dual H-bridge motor drivers.
- Led a **40+ member** engineering organization as Co-President, secured **\$2,500+** in corporate sponsorship to fund custom PCB fabrication, and guided the team to **2nd Place at the 2025 Rowan Regional Competition**.

LPT-KEYPAK

Tewksbury, NJ

Engineering Intern

June 2024 – August 2024

- Commissioned and validated an automated packaging machine for the **Bill & Melinda Gates Foundation**, increasing packaging line throughput by **30%**.
- Diagnosed electromechanical failure points in automated manufacturing equipment, reducing production defect rates by **15%**.
- Interpreted **20+ complex industrial electrical schematics** to troubleshoot PLCs, motor drives, and sensor relays, establishing standardized maintenance SOPs.

VEX Robotics (Rutgers IEEE)

New Brunswick, NJ

Lead Electrical Engineer

September 2024 – Present

- Designed communication and power distribution architecture using the VEX Brain, enabling **RS-485 to UART to I2C** data transfer between embedded subsystems.
- Selected and integrated differential transceivers, **TVS surge protection diodes**, decoupling capacitors, and pull-up/down resistors to protect logic from inductive spikes and signal reflections.
- Conducted functional bench testing and signal analysis on communication lines using digital oscilloscopes to validate signal integrity and reliable data transmission across competition runs.

TECHNICAL PROJECTS

Autonomous Micromouse Robot (Jerrieee)

2024 – Present

C/C++, STM32F405RG, KiCad 8.0, 4-Layer ENIG PCB, DRV8847

- Won **2nd Place at Rowan Regional Competition**; designed custom **4-layer ENIG PCB** integrating **168 MHz ARM Cortex-M4 MCU**, optical ToF sensors over **400 kHz I2C**, and dual **DRV8847** motor drivers in an 84x76.5 mm chassis.
- Programmed real-time Flood-Fill maze navigation algorithm with a **cascaded PID** motion control loop correcting gyro heading and velocity errors with **anti-windup clamping** to maintain stable tracking.

POSIX Robotics Simulation & Telemetry Engine

2025

C++, Qt6, Python, NumPy, SciPy

- Built a headless simulation pipeline in **Qt6/C++** benchmarking pathfinding algorithms across competition maze layouts over POSIX inter-process pipes under Linux.
- Developed a serial telemetry logger capturing motor PWM output, angular error, and ToF distance streams at **100 Hz** using Python and Matplotlib to benchmark PID settling time.